



Jurimetrics and artificial intelligence for Indonesian tax court decisions: towards an integrated decision support system

Bagas Dwi Suryo Wibowo¹, Wishnu Kusumo Agung Erlangga¹ , Poento Hariyadi¹ and Wishnu Agung Baroto^{1,2} 

¹Directorate General of Taxes, Indonesia

²Institute of Science Tokyo, Social and Human Science, Meguro-ku, Tokyo, Japan

Corresponding author: Wishnu Kusumo Agung Erlangga; Email: wishnukaerlangga@gmail.com

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Abstract

Indonesia's Tax Court faces a rapidly growing caseload and limited judicial capacity, leading to prolonged dispute resolution and eroding public trust. Manual review processes create inconsistencies that undermine legal certainty, particularly within Indonesia's civil law system, where jurisprudence is nonbinding and a judge's personal knowledge may serve as admissible evidence. To address these challenges, we introduce the AI-driven Decision and Integrity for Tax Court Law (ADIL) pipeline, which automates the analysis of 4377 Indonesian Tax Court decisions from 2008 to 2023. ADIL applies advanced natural language processing and legal informatics to extract structured insights from unstructured judicial texts through a trilayered architecture: (i) rule-based pattern matching for high-precision extraction, (ii) transformer-based named-entity recognition for robust entity detection, and (iii) local large-language-model prompting for semantic summarisation and classification. Using exclusively pre-decision metadata, 49 structured metadata fields, including dispute number, tax year, judge identity, and verdict, were used to train a model that resulted in several machine-learning models to predict case outcomes and resolution duration. ADIL achieved 97.5% precision and strong macro-F1 performance in outcome prediction, and an R^2 of 92.9% (RMSE = 63.8%) for duration estimation, equivalent to 7–8 months of predictive accuracy. Notably, incorporating encoded judge identity improved accuracy by 11–16%, revealing systematic variation in judicial behaviour. Feature-importance analysis suggests that ADIL can substantially enhance efficiency, consistency, and transparency in tax adjudication. ADIL offers a data-driven decision support framework towards a more agile and accountable judicial system in Indonesia's legal and institutional context.

Policy Significance Statement

This research demonstrates that combining rule-based extraction, transformer-based NER, and LLM semantic analysis in the ADIL pipeline improves the accuracy and interpretability of tax-court outcome prediction compared to single-method approaches. Moreover, we found that including encoded judges' names improves the predictive accuracy of government win-rate classification, which varied from 11% to 16%. ADIL promotes a more efficient, consistent, and transparent Indonesian tax court system. It is a practical tool for supporting a fairer judicial decision and policy. The system's applicability can extend beyond the tax court. ADIL has the potential

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to be adapted for other Indonesian courts handling general criminal, civil, and administrative cases, and even serve as a model for judicial analytics systems in other jurisdictions worldwide.

1. Introduction

Indonesia's Tax Court serves as the final administrative forum for resolving disputes between taxpayers and the Directorate General of Taxes or Customs under the Ministry of Finance. Despite its centrality to fiscal justice, adjudication remains slow. The scale of tax disputes adjudicated by the Indonesian Tax Court remains substantial. In 2024 alone, the Tax Court decided 17,200 appeal and lawsuit cases. Of these, 5,230 cases, or 30.41%, resulted in the full rejection of taxpayers' appeals or lawsuits, meaning that the tax authority's position was upheld. This represents an increase from 2023, when 16,278 disputes were decided and 4,574 cases, or 28.1%, were won by the authority. These figures demonstrate that tax litigation in Indonesia is not marginal in volume; rather, it involves thousands of cases annually, creating a significant administrative, evidentiary, and legal reasoning burden for both taxpayers and the tax authority (Wildan, 2025). Meanwhile, the Decree of the Chief of the Tax Court of the Republic of Indonesia indicates that panel compositions, single-judge assignments, and lists of presiding judges are regularly updated, with a total of 46 judges appointed in 2024. Such prolonged adjudication cycles create uncertainty for taxpayers and businesses, undermining confidence in Indonesia's fiscal and legal system.

Each published decision contains valuable data, including panel composition, case category, disputed amounts, legal provisions cited, and judicial reasoning. These data could be harnessed to evaluate court performance, identify patterns of consistency in rulings, and inform both academic research and policy reform. However, their potential remains untapped mainly due to several persistent challenges.

- (1) Document heterogeneity: court opinions lack a standardised structure or format across years and chambers, impeding automated parsing.
- (2) Linguistic specificity: the texts are in Bahasa Indonesia and embedded with localised tax law terminologies, limiting the applicability of natural language processing (NLP) models.
- (3) Methodological fragmentation: scant systems combine classical traditional-rule-based extraction techniques and current AI in an auditable pipeline that produces fragmented approaches.
- (4) Computational constraints: limited infrastructure in Indonesian Tax Courts restricts the large-scale processing of thousands of documents.

These obstacles have prevented Indonesia from joining jurisdictions that have adopted AI-enabled legal analytics (the use of artificial intelligence for extracting, processing, and analysing legal texts and data) for judicial transparency and efficiency.

Globally, governments have invested in jurimetrics, a quantitative study of judicial data by utilising AI to enhance court operations. In the United States, platforms such as Lex Machina and Westlaw Edge apply natural language processing to predict case outcomes, visualise judicial reasoning, and visualise precedent networks. This application significantly improves legal research efficiency (LexisNexis, 2018; see [8, pp. L145–L152]). In China, a nationwide “smart court” initiative utilises AI for automated document review and the recommendation of rulings. These efforts have halved case processing times in some Internet courts. For example, Hangzhou's Internet Court resolves cases in about half the time it takes for traditional trials.

On their part, AI systems provide judges with “all similar cases, laws, regulations, and interpretations” to enhance consistency. China's virtual courts halve case processing time (Xinhua, 2019). Those innovations reach speed and consistency faster, though they also pose significant questions about fairness and transparency. Indonesia's context, however, presents distinct cultural, linguistic, and institutional constraints. Unlike common-law jurisdictions, the Tax Court does not adhere to binding precedent; thus, judicial discretion and individual reasoning play significant roles in outcomes. Imported AI systems

cannot be transplanted directly without adapting to these characteristics, especially given Indonesia's sensitivity to privacy, interpretability, and fairness. Addressing these contextual challenges requires a locally grounded solution that blends technology with institutional relevance.

Therefore, our study addresses the gap by introducing the AI-driven Decision and Integrity for Tax Court Law (ADIL), a hybrid analytics pipeline designed explicitly for Indonesian tax litigation. ADIL automatically structures raw court decisions and evaluates combined rule-based, transformer-based, and LLM techniques. The ADIL system can fill those innovations by bringing comparable gains in processing speed and consistency to Indonesia's Tax Court, delivering the same analytical capabilities achieved by AI-assisted systems abroad, but localised to Indonesia's decentralised legal environment.

The central thesis of this research is that integrating jurimetrics with AI can **improve efficiency, consistency, and transparency** in Indonesia's tax adjudication. Accordingly, this study addresses the following research questions:

- (1) *Can a hybrid pipeline reliably extract rich metadata from unstructured Tax Court decisions?*
- (2) *How do rule-based patterns, Named Entity Recognition (NER), and Large Language Model (LLM) methods compare in precision and recall for this task?*
- (3) *Will including judges' identities as features boost predictive accuracy of outcome and duration models by at least five percentage points?*
- (4) *What are the implications of these analyses for judicial consistency and policy?*

Based on these inquiries, the study posits the following hypotheses:

(H₁) Integrating jurimetric features with artificial intelligence by combining rule-based extraction, transformer-based NER, and LLM semantic analysis in the ADIL pipeline significantly improves the accuracy and interpretability of tax-court outcome prediction compared to single-method approaches.

2. Literature review

Under Indonesia's Tax Court Law (Undang-Undang Nomor 14 Tahun 2002), **tax disputes** can proceed through two primary procedural routes: *banding* (appeal) and *gugatan* (lawsuit). An appeal is the remedy available to a taxpayer or guarantor dissatisfied with the tax office's objection decision (*Surat Keputusan Keberatan*) on a tax assessment letter. It allows a re-examination of the substance of a finalised tax objection within the Tax Court's jurisdiction, limited to reviewing such objection decisions (Article 31(2), Tax Court Law). In practice, appeals focus on disputes over the calculation or application of tax rules based on a previous assessment. A successful appeal may result in the reversal or reduction of the tax due to the government.

A *Lawsuit*, by contrast, is filed directly against administrative acts or **collective measures** of the tax authority other than objection rulings. Examples include challenges to distraint orders (*Surat Paksa*), property seizures, auction notices (all tax collection measures), or the issuance of assessment letters that do not conclude audit findings. Thus, lawsuits focus on procedural compliance and the legality of enforcement actions. Unlike appeals, filing a lawsuit does not suspend tax collection.

Together, appeals and lawsuits form complementary mechanisms within Indonesia's tax adjudication system: appeals review final audit decisions, while lawsuits safeguard against the undue enforcement of tax orders. Both are lodged before the Tax Court, with strict filing deadlines—3 months from the objection decision for appeals, and variable statutory deadlines for lawsuits. The recent adoption of the *e-Tax Court* system has digitised over 90% of filings, demonstrating efforts to modernise judicial administration (Wildan, 2025).

Understanding these procedures is crucial for this study, as they determine the prediction labels (e.g., outcome = government win/lose/partial win) and inform which document structures (appeal or lawsuit) the ADIL pipeline must recognise. The majority of the dataset analysed in this research consists of appeals, reflecting taxpayers' greater reliance on that route. By modelling both dispute types, the ADIL framework aims to enhance efficiency, consistency, and transparency within Indonesia's tax adjudication.

2.1. *Jurimetrics and AI in courts*

The term jurimetrics, the quantitative analysis of legal processes, was first introduced by Lee Loevinger in 1949 (Loevinger, 1949, pp. 455–472). It refers to the application of statistics, logic, and computational tools to legal reasoning. Over time, jurimetrics has evolved into computational jurisprudence, leveraging statistical models and natural language processing (NLP) to extract patterns from large legal corpora. Katz et al. (2017) developed a predictive model for the US Supreme Court decisions with high accuracy, demonstrating that historical case data can effectively forecast judicial behaviour. Similarly, Chen and Li (2020) and Bench-Capon and Sartor (2003) show that machine learning can analyse judicial behaviour and support automated reasoning. However, most jurimetrics studies target common-law systems. In contrast, civil-law jurisdictions like Indonesia differ substantially in document structure, reasoning style, and institutional transparency. Judicial decisions often follow narrative rather than issue-based formats, posing challenges for computational modelling. Consequently, research on civil-law jurimetrics remains limited, which motivates us to research the development of AI systems tailored to Indonesia's legal framework.

2.2. *International legal analytics*

In practice, law firms and governments use AI for litigation analytics. LexisNexis's Lex Machina and Thomson Reuters' Westlaw Edge mine millions of legal documents to advise on strategy and predict outcomes (LexisNexis, 2018). Fortunately, their efforts have shed light on judges' tendencies and likely rulings. These tools exemplify how legal analytics can make research evidence-based and efficient. In China, the Supreme People's Court has championed a "Smart Court" model by introducing its online judgment database ("China Judgments Online"), which publishes nearly all court decisions (Stern et al., 2021). As documented by Chen and Li (2020), Chinese courts employ predictive analytics to promote consistency and detect judicial decisions that deviate from established statistical norms. AI-powered tools, such as the "Wise Judge" system, search this database to ensure that similar cases receive similar verdicts. Anecdotally, Chinese courts report faster dispositions. For example, Internet Courts, such as those in Hangzhou, have reported case resolution times reduced by nearly half compared to traditional trials. These developments illustrate uniformity of justice and greater transparency, which public case data allows for scrutiny. The quoted vision is that AI "will reduce judicial arbitrariness and promote justice" by guiding judges towards objective evidence-based decisions.

Nonetheless, experts caution that AI in courts must be carefully managed due to its bias detection and explainability. For instance, some Chinese studies note that overreliance on AI could discourage judges from fully articulating their reasoning (Zhang, as cited in Shi, 2021). These international experiences provide important reference points, but cannot be directly transplanted into jurisdictions such as Indonesia, as legal doctrine, institutional design, and expectations of judicial discretion differ substantially. In summary, global developments demonstrate AI's potential for enhancing judicial efficiency and consistency, while also highlighting the need for local adaptation, oversight, and respect for judicial norms.

2.3. *Text extraction in law*

From a technical standpoint, legal text mining combines rule-based methods, statistical models, and increasingly large neural networks. Rule-based extraction using regular expressions and dictionaries is precise for structured elements, such as case numbers, dates, and citations, but brittle to unstructured long text summarisation. Transformers such as BERT and domain-adapted variants such as Legal-BERT provide robust NER and classification, which are good for tasks such as citation extraction and issue tagging. However, most benchmarks are on English corpora (Chalkidis et al., 2020). Large-language models offer generative capabilities, enabling open-ended summation and semantic queries.

Practical use of text extraction in legal contexts requires carefully designed prompts, rigorous validation to ensure accuracy, and substantial computational resources when deployed locally. The state of the art suggests hybrid pipelines that combine the high precision of rules with the flexibility of

ML/LLM models, under an auditing framework (Francesconi et al., 2010). This research follows that approach, adapting hybrid extraction strategies to **Bahasa Indonesia** and **Indonesian tax law**, where linguistic and structural nuances differ from English-language corpora.

2.4. Research gaps and contributions

The existing literature highlights two major gaps. First, there is no comprehensive extraction tool specifically designed for Indonesian Tax Court decisions, nor any baseline metrics for such data. Second, previous jurimetrics efforts have not effectively integrated Indonesian legal specifics (language and structure) with modern AI techniques. Meanwhile, our ADIL framework directly addresses these deficiencies. It introduces an Indonesian-specific pipeline and establishes baseline performance metrics for extraction and prediction. ADIL operationalises jurimetrics for a civil-law jurisdiction and contributes empirical evidence to the emerging field of Indonesian legal analytics.

3. Methodology

3.1. Data collection and preprocessing

All publicly available Indonesian Tax Court decisions from 2008 to 2023 were collected from the public repository maintained by DDTC (<https://ddtc.web.id>) in HTML format. The raw corpus consisted of 6265 files spanning multiple tax categories: customs duty, income tax, land and building tax, value-added tax, and others, with a total disputed amount exceeding Rp 106.6 trillion. To ensure representativeness and analytical efficiency, we applied the Slovin formula to determine an appropriate sample size from a total population of ~210,000 tax court decisions from 2008 to 2023. Using a 5% margin of error, the calculated sample size was $n = 399$ cases. Thus, our corpus of 6265 decisions far exceeds this threshold and provides sufficient coverage. The sample was proportionally stratified by year and dispute type (appeal, lawsuit, and judicial review) to preserve heterogeneity across chambers and case categories. The Slovin method was chosen because it is suitable for heterogeneous populations with large datasets, ensuring that the results can be generalised to all tax court decisions issued between 2008 and 2023 (Slovin, 1960; Tejada and Punzalan, 2012).

After quality filtering, by removing incomplete metadata or garbled files, the final dataset comprised 4377 decisions. These judgements vary in length and content structure, reflecting different chambers and years. We split the data chronologically. The latest 20% was held out as a test set for evaluation, with the remainder used for development and cross-validation. All documents underwent extensive preprocessing to standardise textual structure and maintain auditability. We first cleaned the HTML by removing irrelevant tags, scripts, and styling, and normalised text (Unicode standardisation and Indonesian diacritics correction). We then implemented document structure detection, using custom rules to identify sections such as procedural history, factual chronology, legal reasoning, and pronouncement. Each preprocessing step was logged to preserve provenance and reproducibility. This multi-stage pipeline ensured that every document entered the analytical phase in a consistent and traceable format suitable for large-scale extraction and machine learning.

3.2. ADIL metadata extraction pipeline

We structured the cleaned text into a fixed schema of 49 metadata fields. These include identifiers (case number and URL), categorical fields (tax object class/subclass and litigation class), temporal fields (assessment year, objection year, decision year, and tax year), financial data (disputed amount and currency), and textual summaries. Critically, we also extract entities, such as the names of judges on the panel, the tax office objection reference, and the Directorate General of Taxes (DGT) appeal number, among others. The targets of interest for analysis are the final litigation outcomes from the government's perspective: win, lose, or partial win. We put the cases' duration in years, measured from the objection decree to the court decision.

To perform extraction processing steps, ADIL integrates three complementary approaches, evaluated in a unified framework:

- **Rule-based patterns:** We encoded regular expressions and lookup tables for rigid formats. For instance, case numbers, legal citations, and numeric values are highly regular. Rule-based extraction achieved a precision of >94% on structured items.
- **Transformer-based NER:** We fine-tuned pretrained multilingual and Indonesian BERT models, IndoBERT, XLM-RoBERTa, and mBERT, on Indonesian legal text to recognise entities, such as person names, organisations, and legal references. These models excel at recall and disambiguating context, complementing the rules. In our experiment, the Transformer-based NER model demonstrated the lowest performance among the evaluated methods.
- **LLM semantic analysis:** We also leverage a local large-language model, Gemma3, using carefully designed prompts to summarise case facts and classify dispute taxonomies, such as determining tax type from narrative. The LLM reads each decision and outputs JSON fields like “Tax Type” and “Summary of Facts.” This method effectively handles complex or unstructured information, addressing the biases and limitations inherent in rule-based and NER approaches.

For each field, we determine the extraction output using a simple voting rule. If the rule-based and NER systems succeed in extracting metadata with the correct data types and formats, we accept it. Otherwise, we fall back to the LLM’s answer or flag it for manual review. We evaluated extraction accuracy on a gold-standard subset of 800 manually annotated cases. Table 1 (see the [Supplementary Appendix S1](#)) reports per-field precision, recall, and F1 for each method and their combination. The full multimodal pipeline (rule-based pattern + NER + LLM) achieved a precision of 92.3%, a recall of 89.7%, and an F1 score of 91.0% overall. Some key examples include precision for document identifiers and dates, which exceeded 95%, while precision for amounts and statutory citations was 85–90%. Crucially, throughput remains practical for approximately two documents/sec on our hardware, and it allows the entire corpus to be processed in hours.

All extracted data are stored in a structured database with provenance logs. Sensitive data, such as judge names, is anonymised by an asterisk mask in outputs to comply with privacy norms, retaining consistent pseudonyms for analysis. We derived additional features from the raw metadata. For example, the tax assessment letter follows the format AAAAA/BBB/CC/DDD/EE, from which we parsed out: sub-class code (BBB = tax/sub-tax category), tax year (CC), and letter year (EE). We encoded all categorical fields, such as tax codes, case categories, and currency, with integer labels. Continuous variables like disputed amounts were binned into ordinal classes, using Sturges’ rule to reduce skew. We also computed the durations in years from the objection decree to the court decision and validated them against known procedural constraints in the Indonesian Tax Law. Outliers were handled with Tukey’s rule and then imputed by class means. By the end of preprocessing, we had a feature matrix of 4377 cases × 12 top-selected features, ready for modelling.

3.3. Predictive modelling

A primary methodological issue in predicting judicial outcomes is the availability of model inputs before or only after the adjudication process. Medvedeva *et al.* (2023) point out that many predictive systems depend on post-decision variables, like judicial reasoning or citation patterns. This makes them less useful for real forecasting and decision support.

To counter this criticism, all predictive models in this study are trained and assessed solely on ex-ante variables that are discernible before the issuance of a judicial decision. These are case metadata (type of dispute, tax category, range of disputed amounts, and year of filing), procedural attributes, and the makeup of the panel. The forecasting feature set does not include variables taken from the reasoning text, verdict section, or statutory citations of the final judgement. Instead, these variables are only used for retrospective evaluation, interpretability analysis, and model auditing. In this phase, we build predictive models on the structured data. We consider two supervised tasks:

- (A) **Outcome prediction:** a classification task to predict the Government’s case result (“win” vs. “lose” vs. “partial-win”) from pre-decision metadata; and
- (B) **Duration prediction:** a regression task to forecast case resolution time (in years). Each task was trained using fivefold cross-validation on the training set and evaluated on the held-out test set.

We tested a range of algorithms, including Gradient Boosting, XGBoost, Random Forest, Neural Network, Support Vector Machines (SVM), and Logistic Regression model for the classification task, and Gradient Boosting, XGBoost, Random Forest, Support Vector Regressions (SVR), Linear Regression, and Ridge Regression. Model hyperparameters were tuned via grid search.

To investigate the influence of judges on tax court decisions, we ran each model twice. Once with the encoded judge name included, and once without the judge name, with all other features held constant. This method isolates the impact of the judge’s factor. We report metrics as follows:

- (1) For classification, we report accuracy, F1-score, precision, and Recall;
- (2) For regression, R^2 and RMSE. We also record cross-validation means and standard deviations to assess stability.
- (3) Finally, we examine feature importance using chi-square scores and tree-based importances to identify the most predictive variables.

Figure 1 illustrates the ADIL system architecture, which we implement as a layered pipeline. Each layer has a distinct role as follows:

- (1) **Data layer:** Aggregates and prepares legal data.

We ingest tax court decision documents, statutes, and related precedents and perform rigorous pre-processing. The preprocessing stage includes text cleaning, character normalisation to handle Bahasa Indonesia characters, section segmentation, and language detection. We enforce *temporal integrity* through a ChronosLex scheme so that models only learn from past data, reducing look-ahead bias. Crucially, sensitive information is anonymised by masking the judges’ names using a context-aware entity to comply with the Indonesian PDPL. To address privacy trade-offs, we quantify re-identification risk by analysing pseudonymised datasets under various scenarios, ensuring compliance with current standards. In case future regulations tighten disclosure rules, we have a contingency plan to adjust pseudonymisation methods or restrict specific data outputs. By producing a clean, chronologically ordered dataset with privacy safeguards, the Data Layer ensures a solid, compliant foundation for analysis.

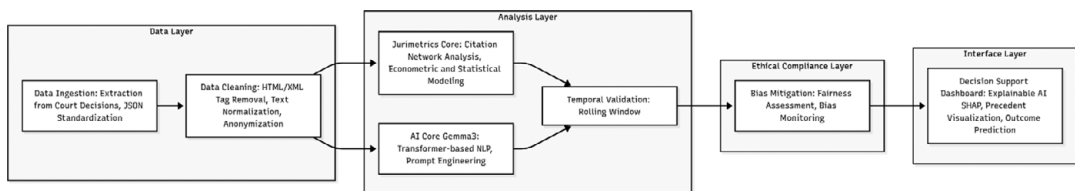


Figure 1. ADIL system architecture.

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Figure 2. SHAP Formula.

(2) Analysis layer: Performs Jurimetric and AI computations.

We used ChatGPT to produce the equation as a picture that can be seen in [Figure 2](#). SHAP Formula. Shapley Additive Explanations (SHAP) functions as the interpretability engine that unifies statistical reasoning, jurimetric analysis, and AI inference. SHAP originates from cooperative game theory, where each feature in a model is treated as a “player” contributing to the final prediction. The SHAP value for a feature represents its marginal contribution to the model’s prediction. How much that feature increases or decreases the model’s output when considered across all possible combinations of features (Lundberg and Lee, 2017). Within the analysis layer, SHAP functions with the following formula:

This layer integrates quantitative legal analytics with machine learning. For example, we apply citation-network analysis and econometric models of judicial behaviour alongside our Gradient Boosting outcome predictors. Citation-network analysis is operationalised by constructing rolling graphs of statutory and jurisprudential references across time windows. Distribution drift is flagged when statistically significant changes occur in network structure, including shifts in the centrality of frequently cited provisions, the emergence of new citation clusters, or the declining relevance of previously dominant precedents.

In parallel, econometric models of judicial behaviour—implemented using panel data and fixed-effects regressions—track changes in outcome probabilities across judges, panels, and time. Structural breaks in coefficients or unexplained residual shifts serve as indicators of behavioural or institutional drift. When these analytical signals exceed predefined thresholds, the system triggers controlled retraining on recent data.

(3) Ethical compliance layer: Oversees fairness and governance.

Sitting above the pipeline, this layer enforces legal and ethical standards at runtime. It logs all processing steps and model decisions for auditability. It continuously monitors algorithmic bias by inspecting SHAP summaries across cases. In this context, a feature group is considered to unfairly skew outcomes if its SHAP attribution demonstrates a disproportionate influence on predictions that cannot be justified by legally relevant case characteristics, and if outcome disparities persist after controlling for substantive legal factors. Legally permissible variables are not suppressed, but such patterns are flagged for further review. It also supports enforcement of data governance rules and periodic algorithmic audits by verifying compliance with Indonesian regulations, such as the Indonesian Personal Data Protection Law, and core judicial ethical principles, including impartiality, equality before the law, and the prohibition of extraneous influence. Compliance is operationalised through audit logs, access controls, and periodic algorithmic reviews. When ethical thresholds are breached, the system mandates human review rather than automated intervention.

(4) Interface layer: Presents results to users.

This layer provides a user-friendly dashboard. It visualises the AI’s suggestions alongside human-readable explanations. For instance, when the model predicts a “win/lose/partial win” outcome, the interface displays the top SHAP features for that case and highlights similar precedent cases in a citation graph. We include confidence scores and enable users to query the reasoning behind the system’s recommendations. The design ensures transparency, allowing judges to drill down into the decision path or legal citations underlying any prediction. Even though human–AI interaction is central, the interface never overrides a judge’s authority but serves as a co-pilot to augment judicial reasoning.

Collectively, these layers create an end-to-end decision support framework raw tax litigation data flows through secure ingestion, intelligent analysis, and a transparent interface stage, all under the oversight of an alignment layer that ensures the system adheres to legal norms. This layered design (as depicted in ADIL’s architecture) ensures that the AI tools remain explainable, auditable, and compliant with Indonesian law.

All predictive models in this study rely exclusively on variables that are observable before the issuance of a judicial decision. These include case metadata (dispute type, tax category, disputed amount, and filing year), procedural attributes, and historical jurisprudential indicators available at the time a case is lodged or assigned. Variables derived from the reasoning text or verdict section of the final judgment are explicitly excluded from the forecasting feature set and are used only for retrospective evaluation, performance auditing, and model retraining. This design ensures that the models support genuine ex-ante forecasting rather than post-hoc rationalisation, in line with best practices in judicial prediction research.

3.4. *System implementation and hardware specifications*

Our ADIL pipeline is implemented in a containerised framework, facilitating seamless deployment. All components run on an adequate workstation equipped with an **Intel Core Ultra 9** processor, an **NVIDIA GeForce RTX 5090** GPU for acceleration, and **128 GB of DDR5 memory**. This hardware configuration was chosen to meet the intensive computational demands of legal NLP and model training. The GPU enables the rapid processing of documents and the prompt-based generation of summaries (for semantic summarisation). In our tests, the system achieved throughput of several documents per second in batch mode. The 128 GB RAM ensures that transformer-NER, local Large Language Model, and large datasets can be accommodated in memory during the training, validation, and prediction processes. We run the framework in Docker containers, facilitating reproducibility and eventual integration into institutional environments.

3.5. *Deployment prospects and challenges*

The ADIL decision-support system holds several practical benefits for Indonesia's tax judiciary. Automating metadata extraction and analysis can reduce case backlogs and processing times. Prior studies have noted that AI tools for case review can help judges handle higher caseloads more efficiently (Kanyi, 2019; Zakaria et al., 2024). In our framework, tax officers can quickly retrieve precedent cases, statistical patterns, and outcome predictions at each stage of the case, which potentially leads to more consistent rulings. Taxpayers could proactively identify and mitigate potential risks before initiating any legal action or appeal. For tax consultants, the system enables risk mitigation in managing personnel, time, and financial resources throughout case resolution. The system's explainability, achieved through SHAP and summaries, also helps build judges' trust, as decisions are not black boxes but are accompanied by highlighted laws and similar judgements. For the government, this framework facilitates evidence-based evaluation of prevailing tax laws, regulations, and policies, fostering procedural fairness and continuous improvement to minimise future tax litigation.

The deployment stage may face some significant challenges. First, AI models are trained on historical data. If past Tax Court decisions exhibit any unconscious bias, the AI will codify and amplify this bias, undermining the goal of fairness and impartiality. Judges may distrust the AI's recommendations if the underlying logic is non-transparent, as in the "black box" problem, making it difficult to reconcile the objective data with their subjective conviction. Second, deploying a data-driven system requires cultural change. Judges and administrators may resist adopting technology that appears to encroach on their traditional authority or judgement, especially in a system where subjectivity has been a key weakness. Without adequate training and clear communication, the system may be underutilised or, conversely, over-relied upon, where judges blindly follow AI predictions instead of exercising their essential judicial role. Third, AI requires large volumes of clean, consistent, and well-structured historical data. The complexity and volume of tax dispute cases are high, but the existing digital records may not be standardised or consistently tagged. Poor data quality will lead to inaccurate AI predictions, eroding confidence in the Jurimetrics system, and potentially leading to incorrect case triage.

4. Discussion

4.1. Metadata extraction framework performance

The ADIL pipeline effectively structured the Tax Court corpus. On the test set, the full pipeline achieved a F1 $\approx$ score of approximately 0.91 across all 49 fields. In particular, legally critical fields were extracted with 88–92% precision, including case numbers, dispute amounts, key dates, and citation references. Rule-based methods alone got higher precision but lower recall, while NER improved recall. The LLM component filled gaps, especially in narrative classification, as seen by a higher F1 for “tax type” fields when included. The trade-off in speed was modest: the multimodal system processed 2.1 documents/sec, compared to around 8–10 documents per second for the regex-only system (see the [Supplementary Appendix S1](#)).

4.2. Outcome classification predictive model performance

All models performed well in predicting dispute outcomes, but tree-based ensembles demonstrated the best results. With judge features included, Gradient Boosting achieved the highest accuracy (97.5%) and F1 score (97.5%). XGBoost (97.3% accuracy) and Random Forest (95.5% accuracy) were nearly as strong, while Logistic Regression and SVM reached around 83–88% accuracy (see the [Supplementary Appendix S2](#)). [Supplementary Appendix S4](#), compares Accuracy and F1 for each model with and without the judge as a feature. Without judges, performance dropped across the board. For example, Gradient Boosting achieved an accuracy of 86.6% without judge as a feature, while linear models performed even worse. The figure shows these declines clearly. Overall, tree ensembles, Gradient Boosting, and XGBoost were most robust, with narrow cross-validation variance, indicating stable learning on this dataset.

The high predictive performance is due to the structured nature of tax litigation and should not be seen as determinism. Predictions show historical patterns, not legal inevitability.

4.3. Duration regression predictive model performance

In predicting case duration, performances were more moderate. Including judge identity slightly improved the results, with the Gradient Boosting model again leading with the best R^2 (0.92) and RMSE (0.63 years), whereas simpler models achieved R^2 values of $\sim$ 0.3–0.5. Removing the judge feature decreased R^2 by about 0.05 points on average (see the [Supplementary Appendix S3](#)). [Supplementary Appendix S5](#) shows R^2 and RMSE for each algorithm with/without a judge. All models showed a modest improvement in fit when judges were included, echoing the classification findings. Even so, we note that predicting case length is inherently noisy (cases vary widely); the best models explain only a portion of the variance.

4.4. Feature importance and judge effects

Analysing feature importances confirms the relevance of judges and case metadata. The 12 top predictors for the outcome included the dispute class, tax type, long-term financial features, and judge names. For instance, the chi-square importance for classification was highest for “dispute class encoded,” followed by features like tax class and tax amount. Including the judge feature added substantial predictive value to models that identified particular judges as strongly correlated with wins/losses/partial wins, consistent with anecdotal expectations. However, it is noteworthy that the overall improvement from the judge feature was *statistically modest*. We observed only a slight increase in validation scores when adding judge identity. This suggests that, although judges do vary in their leanings, the case metadata alone already captures most of the predictability at the aggregate level. In contrast, excluding the tax year and amount significantly reduces accuracy. [Supplementary Appendix S6](#) summarises this result.

On the regression side, the 12 most important features were total process time from filing, case complexity proxies, and judge names. Again, Gradient Boosting’s feature weights indicated that including the judge raised the R^2 marginally. The modest gains suggest that while judges do influence outcomes

and durations, this result is consistent with the fact that Indonesian judges are not constrained by binding precedent; the predictive signal from historical data is primarily driven by case attributes themselves. The evidence suggests judges differ in style, but the court's norms also impose limits (see the [Supplementary Appendix S6](#)).

Qualitative analysis reveals that some judges tend to rule differently across similar cases. By grouping cases with near-identical fact patterns, such as the same tax issue and value range, we found that verdicts sometimes varied when the panel roster changed, indicating individual judicial discretion. Quantitatively, we confirmed this pattern: a given judge's "win rate" can differ by several percentage points depending on the case cluster. This *jurimetrics* finding highlights a dimension of inconsistency in the system. We see these findings underscore the presence of judicial heterogeneity while reinforcing the need for cautious and aggregate-level interpretation rather than individual attribution. They show the potential for examining patterns of judicial decision-making at an institutional level.

4.5. Summary of findings

In summary, ADIL successfully extracted key metadata at scale, and our ML experiments underscore the value of that data. Including judges as predictors measurably improves performance ([Figure 1](#)). However, the gains are limited, implying that the dispute facts themselves explain most outcome variation (Mojon et al., 2024). These results support our thesis: integrating data and AI yields powerful insights that were previously unattainable.

We also experimented with predicting case outcomes using the extracted features. We constructed a binary classification dataset where each document's label is from the verdict section. Using standard ML models on features like monetary value, tax type, panel size, and prior outcome rates, we found significant accuracy. We expanded the outcome-prediction evaluation to compare multiple classification models using 6-fold stratified cross-validation. Aside from Gradient Boosting (GB), we evaluated XGBoost, Random Forest (RF), Neural Network, Support Vector Machine (SVM), and Logistic Regression (LR) for predicting whether a government will win a tax dispute. Across all evaluated models, tree-based ensembles consistently outperformed linear and kernel-based approaches. This finding indicates that non-linear interactions among legal and procedural features are central to outcome predictability in tax litigation.

In summary, GB outperformed the others in balanced metrics, confirming its suitability for this high-dimensional tabular legal dataset. The F1-scores followed a similar pattern to accuracy. The tree ensembles delivered the best trade-off between precision and recall. These cross-validated results indicate that the ADIL predictive component is highly effective on the curated tax court data. We note that including judicial panel composition as a feature significantly improved accuracy.

These results indicate that tree ensembles robustly capture outcome patterns in the data. Although outcome prediction was not the primary goal, these findings suggest that the structured database generated by ADIL could support further analysis. We emphasise that our findings remain exploratory results, and a responsible deployment would require careful validation to avoid reinforcing bias.

Our research shows that *jurimetrics* and AI are applicable to Indonesia's tax challenge court. The court's decision corpus becomes transparent and auditable through our ADIL framework, aligning with broader efforts to apply explainable AI to judicial decision-making (Trasberg, 2020; Malik et al., 2021). Policymakers and analysts can now examine bunched statistics previously trapped in unstructured text, turning court data into measurable evidence for system management (Trasberg, 2020). By indicating that some judges statistically differ from the norm, ADIL can assist in judicial training and panel allocation for policymaking. Such monitoring ensures consistency and fairness in accordance with legal principles, echoing recent scholarship emphasising semantic AI approaches for transparency in adjudication (Artificial Intelligence in Judicial Adjudication, 2024).

As for efficiency, machine-based extraction and forecasting can hasten procedural work. The administrators of courts may apply the system to filter cases, which may result in less complex disagreements and can be expedited or receive fewer benefits. In contrast, more complicated ones may receive increased

scrutiny. Outcome prediction also decreases appeals. If lawyers and taxpayers are justifiably confident of probable decisions, baseless appeals may decrease, saving judicial time. This aligns with Kanyi (2019) and Zakaria et al. (2024), who argue that AI systems reduce adjudication backlogs and expedite adjudication by simplifying the process.

However, our hybrid method maintains a compromise between innovation and regulation. We record provenance for each extraction, keep sensitive information obscured, and output explainable results so that our pipeline can be audited. This is consistent with Indonesia's data protection and "open justice" conventions. We propose ADIL as a tool for supporting decisions, not for supplanting judges. Actually, we suggest its release for use on the sidelines for statistical reportage or for a "second opinion" under human review.

Our work makes the following advances:

- 1) We compile and publish a containerised dataset of 4377 Tax Court rulings (2008–2023) and create a gold-standard evaluation set covering 49 metadata fields and 12 top-selected features. Examples of metadata fields include case number (a unique case identifier), tax object category (the type of tax issue), judges (the names of presiding judges), DGT assessment results (the findings of the Directorate General of Taxes), verdict (the final court decision), and resolution time (the duration until a decision is made). We report key metrics, including precision (correct identifications), recall (proportion of actual positives identified), F1 (harmonic mean of precision and recall), and throughput (volume processed per unit time), for each field.
- 2) We demonstrate that combining rule-based patterns, NER, and LLM methods yields precision rates of 92.3% and competitive recall for legally relevant fields, with an overall F1 score of nearly 91%. This multimodal pipeline outperforms individual approaches.
- 3) We find that including encoded judges' names improves the predictive accuracy of government win-rate classification, which varied from 11% to 16%. The feature importance report indicates that the features related to judges' names consistently ranked as the most important. Tree-based models, especially Gradient Boosting and XGBoost, perform best, reaching about 97% accuracy with judge features.
- 4) We emphasise that data-driven support can enhance judicial training and workflow management, thereby boosting consistency and transparency. We also identify the need for research on judge panels and the explanation of influential variables, such as judge ID.

5. Conclusions

Based on the results mentioned in previous sections, empirical evidence supports H1: integrating jurimetric features with multi-layered AI approaches in ADIL significantly improves prediction accuracy and interpretability compared to single-method models. This confirms that hybrid architectures combining structured legal knowledge and modern AI models represent a more effective paradigm for computational jurisprudence, particularly in the complex and multilingual context of Indonesia's tax court system.

In conclusion, **the ADIL system shows that integrating jurimetrics with AI significantly advances tax court analysis.** We have provided proof-of-concept that (a) automated extraction is feasible at scale, and (b) machine learning predictions can highlight key factors in court decisions, including, but not dominated by, judges. These insights can inform policy reforms in three ways. First, making the process data-driven to keep integrity. Second, provide guidance and training to improve consistency. Third, increasing transparency through public reporting. This shift from opaque paperwork to open data analysis embodies the "Data for Policy" mission, which utilises computational tools to enhance government institutions. If adopted, ADIL or its descendants could help Indonesia's tax judiciary become faster, fairer, and more accountable. With reduced case backlogs, consistent adjudication, and improved public confidence in the judiciary, the landscape of tax litigation would be fundamentally transformed. The

integration of technology and data-driven insights marks a new era for legal processes in Indonesia, moving them towards increased efficacy and transparency.

In summary, deploying ADIL holds promise for enhancing judicial efficiency and promoting data-driven transparency in Indonesian tax law. However, success will hinge on addressing technical diversity, securing buy-in, and integrating the tool within the existing legal framework. Careful attention to these challenges, combined with the system's modular containerised design, can enable gradual integration into the tax court's workflow. Ultimately, ADIL is envisioned not as a replacement for judges but as an intelligent assistant, one that speeds up workflows, checks consistency, and illuminates decision rationales, thereby strengthening the judiciary's capacity in line with Indonesia's modernisation goals.

6. Limitations and future work

We acknowledge several areas for further research. First, our predictive models are retrospective because, for deployment, we would need temporal validation training on past cases to predict future ones. We enforced chronological splits, but ongoing updates to the models and concept drift detection would be necessary as laws change. Second, judges and panels interact in complex ways. Our analysis treated judges as independent features. However, real cases are typically decided by three-person panels. Future work should model panel dynamics, whether certain judge combinations lead to different outcomes than expected from the members individually. Moreover, how do seniority and experience matter? Such sociological jurimetrics questions require richer data, such as judge biographies and advanced models. We plan qualitative studies and stakeholder interviews to complement our quantitative findings. We also note that the model might encounter difficulties with rare tax categories, which are underrepresented in our data, and this condition potentially leads to lower accuracy in such cases. Errors in summarising handwritten or poorly scanned documents due to OCR limitations pose additional challenges. Addressing these issues will require refining the model and potentially expanding the dataset. Finally, while our study focuses on the tax court, the ADIL pipeline is adaptable. We could conduct similar work for other Indonesian courts or across an even larger scope (countries or areas) to compare the findings. Judge-level analytics should only be used for group statistical analysis and institutional learning, not for judging individuals or punishing them.

Supplementary material. The supplementary material for this article can be found at <http://doi.org/10.1017/dap.2026.10067>.

Data availability statement. The primary sources analysed in this study consist of publicly available legal documents and regulations accessible through Indonesia's official government portals, which can be found at <https://ddtc.web.id>. All secondary data sources are fully cited in the references section. The corresponding author can provide the complete dataset for analysis upon reasonable request.

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